03-MECHANICAL

LOT Pager — Mechanical & Industrial Design

Form

- 40 × 40 mm square, corner radius 4 mm. Two **316L stainless** shells.
- **Back shell: mirror polished** (the “flat silver” face). **Front shell: satin**, with glass window carrying screen, camera pinhole and the COPY button.
- Seam: 0.8 mm PEEK/epoxy band = Wi-Fi antenna slot + water seal.

Height stack-up (mm)

Layer	v0	v1	Stretch (5 mm)
Back shell floor	1.0	0.8	0.5
Battery	3.0	2.5	1.2
PCB (0.8) + tallest part	1.4	1.2	1.0
Qi coil + ferrite (face-glass side)	0.7	0.5	0.4
Display module	1.8	1.6	1.0
Camera (Z)	3.5 *(beside screen, not stacked)*	3.0	1.5
Glass	0.5	0.5	0.4
Total	**≈ 9.0**	**≈ 7.0**	**≈ 5.0 (≈15 mAh)**

Camera sits beside the screen in X–Y, so it doesn’t add to stacked height beyond its own Z.

Face layout (front, 40×40)

Screen 27×15 mm top; camera 4 mm lower-left; COPY button lower-right (3×3 actuator under 0.3 mm flexure or small domed key); privacy LED beside lens.

Wireless charging

Qi coil at front-glass side with ferrite toward PCB. Device charges **front-down**. Charger pad has a 40×40×1 mm pocket + magnets (N52 ⌀3×1 ×4) for alignment. Keep steel ≥ 2 mm outside the coil radius or expect large loss.

Finishing

Back: 316L, electropolish + mirror (target Ra < 0.05 µm). Front: bead-blast satin. Optional anti-fingerprint PVD on the back.

Files to create next

STEP model, DXF outlines for PCB (PCBWay), PCBWay CNC drawing PDF with tolerances (±0.05 mm).